

Biliary Leaks: Multidisciplinary Approach to Diagnosis and Treatment

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Bile leaks arise from various causes such as trauma, complications after hepatobiliary surgery, and intrahepatic malignancies or their associated liver-directed treatments. Bile leaks can result in significant morbidity and mortality. Delayed diagnosis is not uncommon due to nonspecific manifestations; therefore, a high clinical suspicion is needed. A multidisciplinary approach for treatment of biliary leaks with prompt referral to tertiary care centers with experienced hepatobiliary surgeons, advanced endoscopists, and interventional radiologists is needed to address these challenging complications. Management of biliary leaks can range from conservative management to open surgical repair. Minimally invasive procedures play a crucial role in biliary leak treatment, and the interventional radiologist can help guide appropriate management on the basis of a clear understanding of the pathophysiology of biliary leaks and a current knowledge of the armamentarium of treatment options. In most cases, a simple diversion of bile to decompress the biliary system may prove effective. However, persistent and high-output biliary leaks require delineation of the source with tailored treatment options to control the leak. This may be done by additional diversions, occluding the source, reestablishing connections, or using a combination of therapies to bridge to more definitive surgical interventions. The authors describe the different treatment options and emphasize the role of interventional radiology.

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Abbreviations: CBD = common bile duct, ERCP = endoscopic retrograde cholangiopancreatography, HIDA = hepatobiliary iminodiacetic, NBCA = *N*-butyl cyanoacrylate, PTBD = percutaneous transhepatic biliary drainage, PTC = percutaneous transhepatic cholangiography, TACE = transarterial chemoembolization

TEACHING POINTS

- A persistent and high output of bilious material after percutaneous drainage of a biloma or any collection is indicative of a biliary leak and/or fistula.
- Drains may stay in place until the patient's clinical picture improves, drainage output decreases significantly (<10 mL over 24 hours), or follow-up imaging demonstrates a resolution or significant reduction in the size of the biloma. A transhepatic drain may be required to stay in place for up to 4–6 weeks to allow the tract to mature.
- Injecting existing surgical or percutaneous drains, when present, can help to opacify the bile ducts and identify the source of the leak.
- Embolization of biliary fistulas is usually reserved for leaks that are not controlled with diversion after PTBD or endoscopic biliary stent placement.
- The use of covered self-expanding metal stents (SEMS) has the dual purpose of sealing the leaks and treating the disruption or stenosis of the bile ducts.

Introduction

Biliary leaks are devastating iatrogenic complications that can occur after hepatobiliary surgery and liver-directed interventions, resulting in significant morbidity and mortality rates, an impaired quality of life, and financial burdens for affected patients (1). These complications are caused by damage to the bile ducts, unsuccessful bile duct ligation, or failure to recognize anatomic variants during surgical interventions. In addition to iatrogenic causes, biliary leaks can be secondary to traumatic or neoplastic processes. Early recognition of a biliary leak is essential to assist in treatment and management because biliary leaks can be devastating. Unfortunately, many biliary leaks are not identified at the patient's initial presentation, with many patients having symptoms of vague abdominal pain and fever similar to those of other postoperative sources of pain.

The leading cause of biliary leaks across the world is hepatobiliary surgery, particularly surgical removal of the gallbladder. Postoperative leaks have a prevalence of 1% for laparoscopic cholecystectomies and 0.5% for open cholecystectomies (1–4). There was an increase in biliary complications after the introduction of laparoscopic cholecystectomies, likely secondary to the learning curve associated with laparoscopic surgery, but this has decreased in recent decades (3,5). Other hepatobiliary surgeries including hepatectomy, living donor liver donation, and liver transplantation can be complicated by biliary leaks. Bile leaks can also result from catheter retraction or dislodgement after percutaneous transhepatic biliary drainage (PTBD). In addition, biliary leaks are uncommon complications of liver biopsies and liver-directed therapies such as transarterial chemoembolization (TACE) and thermal ablation (6). These complications remain an important differential consideration for any invasive

interventional radiation therapy, and with modern interventional techniques and awareness of these complications, it is predicted that the rate of biliary complications will decrease.

Most biliary leaks manifest clinically as bilomas, with symptoms that range from asymptomatic and nonspecific abdominal pain to sepsis due to a superimposed infection (1,2). Bilomas can manifest as well-circumscribed collections external to the biliary tree (Fig 1). In the abdomen, biliary leaks can result in intra- or extrahepatic bilomas (3). Free spillage of bile into the peritoneum can result in bilious ascites with or without peritonitis. Chest involvement can manifest as bilious effusions (bilothorax) or even bronchial fistula and can lead to more serious respiratory complications such as pneumonia, empyema, and adult respiratory distress syndrome (7) (Fig 2). Biliary leaks can also result in biliary fistulas between the biliary tree and the skin or other luminal cavities (8).

There is no standardized algorithmic approach to the diagnosis and treatment of biliary leaks, although many have been suggested (9). After referral to the interventional radiologist for evaluation and management, there are many therapeutic approaches to treat biliary leaks, including a general stepwise approach and the ability to increase invasiveness as clinically appropriate.

Etiology, Prevalence, and Pathophysiology

Cholecystectomies, particularly with the laparoscopic approach, are the most common cause of biliary injuries. Concerted efforts to prevent postcholecystectomy bile leaks have led to a prevalence of approximately 0.15%–0.46% (4,9–14). Given this estimate, cholecystectomies can contribute to several thousand cases of biliary leaks per year in the United States (12,13). Patients diagnosed with a bile leak complication have a twofold increase in the 1-year mortality ratio (13). The most commonly reported reasons for biliary leaks after laparoscopic cholecystectomy are slippage of the cystic duct ligature, gallbladder bed leakage from hepatic parenchymal injury, and anomalous biliary duct anatomy, such as an unrecognized duct of Luschka or other aberrant ducts (1,4,9).

Hepatic resections contribute to fewer annual cases of bile leaks. However, bile leaks are the most reported postprocedural complication from this surgery, with rates ranging from 3.6% to 20% (6,8,9,14–16). Predictors of high-risk resections for bile leaks have been identified and include peripheral cholangiocarcinoma, extended left hepatectomies, resection that involves the main portal fissure, and central resections (16). Hydatid cyst resections appear to pose the highest risk of surgical intervention for biliary leaks and have a poor postoperative prognosis (17).

Nonsurgical causes of biliary leaks include trauma, underlying liver disease, and associated liver-directed therapies. In the setting of blunt liver trauma, the incidence of biliary leaks is less than 5% (6,18–20). Perihilar liver injuries may be associated with high-output biliary leaks that may require additional interventions, while more peripheral injuries tend to be self-limited with conservative management (20).

Biliary leaks are an uncommon complication of liver-directed interventions such as percutaneous biopsy, TACE, and thermal ablation (21,22). In one study, a higher incidence

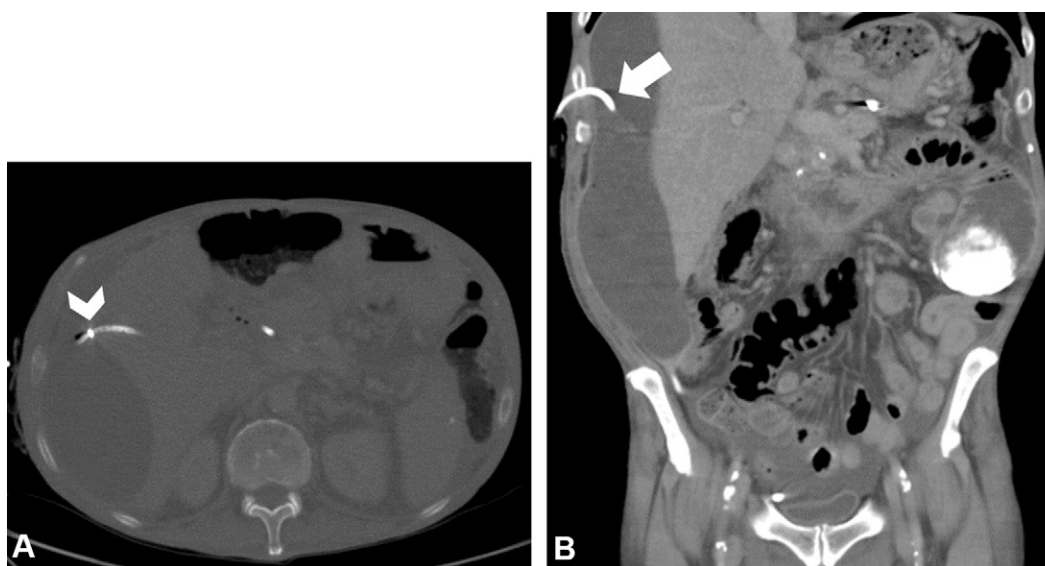


Figure 1. Biloma in a male patient. Axial (A) and coronal (B) CT images show a large extrahepatic biloma secondary to PTBD (arrow in B) and proximal side hole migration outside of the liver (arrowhead in A).

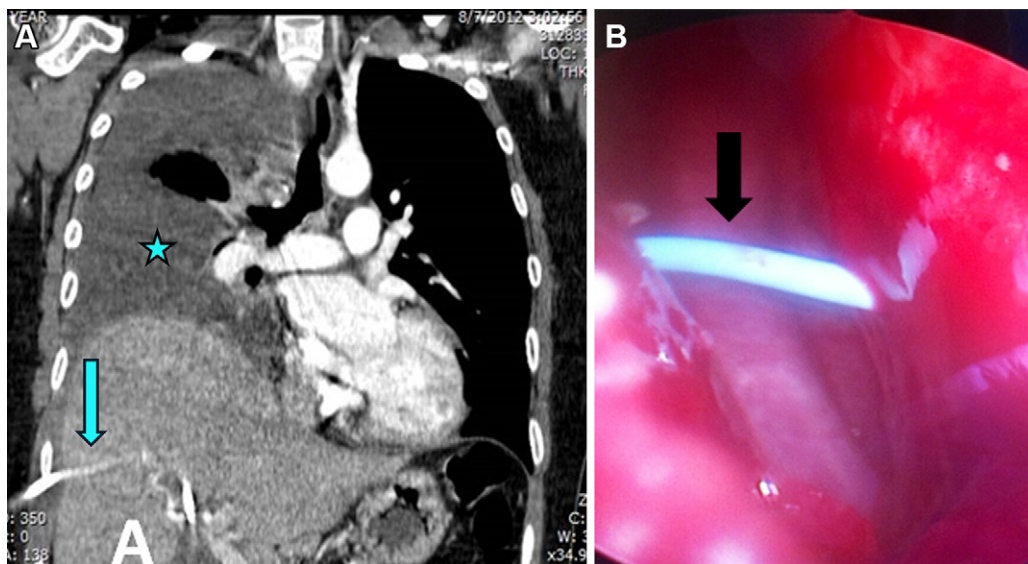


Figure 2. Bilothorax in a 44-year-old woman who underwent complicated laparoscopic cholecystectomy that required hepaticojejunostomy. The patient developed cholangitis that required an internal-external drain due to biliary obstruction. Coronal contrast-enhanced CT scan (A) and photograph during video-assisted thoracoscopy (B) demonstrate a large bilothorax (★) after the internal-external drain (arrow in A) accidentally crossed the pleura. Note the drain crossing the pleural recess in the photograph (arrow in B).

of bile duct injuries was reported with use of drug-eluting beads (DEB)-TACE compared with that for conventional TACE (0.4%–4%) (21,22). In the past, a remarkably high incidence of bilomas of 54% was observed in seven of 13 patients with neuroendocrine tumors metastatic to the liver that were treated with DEB-TACE (23). More recent studies have not demonstrated a higher incidence of biliary injury after DEB-TACE, with rates reported at 0.87%–9% for biliary injury and 0.8% for biloma formation (24–26). The Society of Interventional Radiology quality improvement guidelines currently recommend a 2% incidence threshold after TACE biloma for review of practice patterns (27). Arterial embolization can result in ischemic-induced necrosis and subsequent stenosis of the more central bile ducts, given their nearly exclusive dependence on arterial blood supply (22,26). The result can be biliary dilatation with stasis or bilomas, which carry a risk for infection or, in rarer cases, can expand and rupture.

Biliary tract damage after liver ablation has been reported, with an incidence as low as 0.1% and as high as 12%, depending on the location and ablation modality (3,28). Bile injuries may result from damage caused by the needle, direct thermal injury, or damage to the microvasculature of the biliary tract (28,29). Please see the “Complications” section for further information. Risk factors for biliary injury after ablation have been reported and include a history of prior TACE and lesions located close (<10 mm) to the main biliary ducts (28,29).

Infected bilomas and biliary leaks after liver-directed therapies are particularly common in patients with a history of previous bilioenteric anastomosis and biliary stent placement and can result in severe complications including sepsis, liver failure, or death (6,30).

Understanding the pathophysiology related to biliary leaks can help direct diagnosis and subsequent management. Surgically associated biliary leaks often occur in the presence of

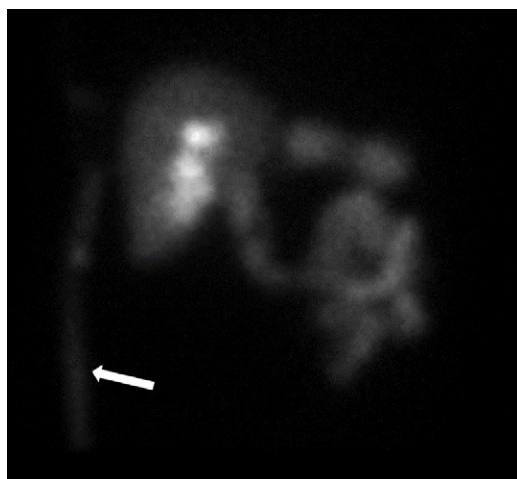


Figure 3. Cholecystitis in a 67-year-old man who underwent laparoscopic cholecystectomy and presented with a biliary leak. Coronal ^{99m}Tc HIDA cholescintigraphy image shows leakage of radiolabeled material entering a drain (arrow).

a pressure differential. Bile, similar to any fluid, will take the path of least resistance, resulting in spillage of bile. Depending on the location and size of the defect, the flow and leakage of bile will vary. Injuries to the smaller parenchymal biliary ducts will have a lower flow rate than injuries to the central biliary system, with peripheral lesions being more difficult to isolate.

Elevated distal biliary pressure may also be caused by stenosis from an extrinsic compressing mass or a surgical clip across the common bile duct (CBD) if the duct is accidentally mistaken for the cystic duct during surgery. Hence, a diversion through a low-pressure system, such as a stent, facilitates closure and healing of the biliary injury (10,18,19). Biliary stents that cross the sphincter of Oddi keep the sphincter open, decompressing the entire biliary tree. A similar decompression can also be achieved by placement of an internal-external percutaneous biliary drainage catheter.

Diagnosis

In approximately 25%–32% of cases, an iatrogenic bile duct injury is recognized intraoperatively (1,6,10), and these biliary injuries may be repaired by the surgeons during the initial operation if they have sufficient hepatobiliary surgical experience, or the bile leak can be drained by placement of a right upper quadrant surgical drain with additional prompt transfer of the patient to a tertiary care center (10).

Depending on the location of the biliary leak and the presence of suspected superimposed infection, patients with biliary leaks can present clinically with acute symptoms such as sepsis, peritonitis, pain, fever, or high output of bilious fluid through or around indwelling surgical drains or surgical wounds (3,6,9,10). If the bile leak is slow, there is often time for fibrosis and encapsulation to occur, followed by formation of a biloma. If the leak is rapid, biliary peritonitis is more likely to occur (3). Jaundice may or may not be present (10,18).

In the proper clinical setting, such as a procedure performed near the biliary ducts, the clinician should maintain a high suspicion for postintervention biliary leaks. However, because biliary leaks may occur weeks, months, or years later (1,6,9,31), the clinician should still include a bile injury in the differential diagnosis of potential complications.

Laboratory findings tend to be nonspecific with elevated serum markers, such as high white blood cell count for a superimposed infection in the bilomas. Elevation of the conjugated bilirubin level in serum with no other elevation of the liver enzyme levels is frequently seen, although this may vary because alkaline phosphatase, aspartate transferase, or alanine transferase levels will occasionally also be elevated (3,10,18). The most confirmatory laboratory test for a biliary leak is a marked elevation of bilirubin levels in fluid aspirate. In cases of suspected bilothorax, a pleural bilirubin-to-serum bilirubin ratio of greater than 1:1 is diagnostic (7).

The clinical history of a patient is often used to suggest a leakage site when not readily apparent at imaging. Collections around a biopsy tract are suggestive of an intrahepatic location, while collections adjacent to the gallbladder bed after biliary surgery are suggestive of an extrahepatic leak. Localization of the biliary leaks, specifically whether the leak is intra- or extrahepatic, is a key component of treatment planning.

Diagnostic Imaging

Diagnostic imaging may be indicated after clinical suspicion for a bile leak is proposed, and the radiologist should use the patient's clinical history to integrate this differential consideration. US is often ordered during the postsurgical period, particularly after hepatectomy or liver transplantation, to allow assessment of the liver, vasculature, and perihepatic fluid collections. Bilomas may manifest as anechoic and well-circumscribed collections. Complex collections with internal septa can be suggestive of a superimposed infection (31).

CT is frequently performed when there is a higher clinical suspicion for complications or if the patient is more acutely ill. CT findings can be nonspecific, often demonstrating only perihepatic collections (3,6,9,10,31) (Fig 1). When fluid collections are present, the thickness of the rim may suggest chronicity, with thicker rims suggesting a more chronic process (3,31). The attenuation of fluid on CT images can be helpful because bilomas are commonly less than 20 HU (3,31). Overall, these findings are nonspecific, and the differential diagnoses include hematoma, seroma, abscess, cyst, and lymphocele (1,3,10,31). An advantage of CT, in addition to its rapid availability, is the ability to depict the anatomy and other associated findings such as peritoneal thickening or enhancement suggestive of biliary peritonitis (3,9,10,31).

Cholescintigraphy with technetium ^{99m}Tc hepatobiliary iminodiacetic acid (HIDA) provides physiologic and functional information. This procedure involves use of ^{99m}Tc HIDA as a radiotracer, which is excreted into the biliary tree by hepatocytes. This modality may also be used to help assess the presence of radiotracer in abdominal drains, which suggests bile leak (3) (Fig 3). Unfortunately, cholescintigraphy lacks spatial resolution and can be time intensive, taking

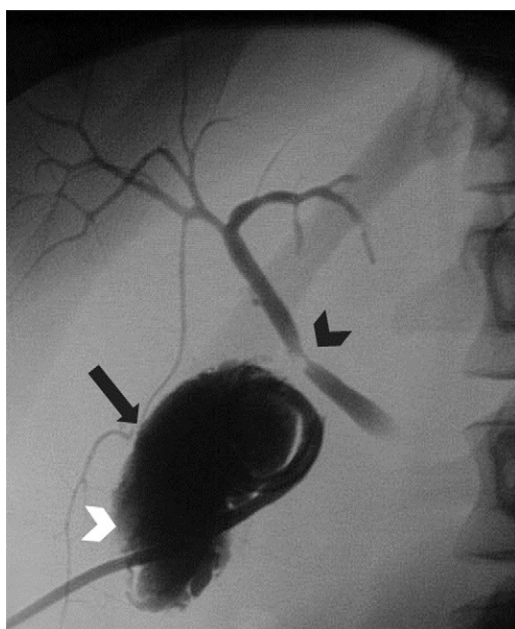


Figure 4. Biloma in a man who underwent laparoscopic cholecystectomy and percutaneous drainage. Coronal cholangiogram shows biloma formation in the gallbladder fossa surgical bed (white arrowhead), an injury of a small aberrant duct (arrow) in segment 6, and an additional incomplete CBD transection (black arrowhead).

4 hours or up to 24 hours when leaks are slow (3,31). Since many leaks are too small to allow localization of the injury with CT alone, imaging with a higher spatial resolution may be required. Spectroscopy CT can improve localization of biliary leaks in relation to the liver capsule and parenchyma by combining planar imaging with anatomic detail (3,6,9,31). A focal collection of radiotracer accumulation is a suspicious finding when CT findings confirm a fluid collection. Other useful options to improve anatomic visualization include MR cholangiopancreatography (MRCP), endoscopic retrograde cholangiopancreatography (ERCP), or percutaneous transhepatic cholangiography (PTC).

Abdominal MRI and MRCP can provide further details of the fluid characteristics and allow further anatomic assessment of the biliary tree. Bilomas classically are hypointense on T1-weighted images and hyperintense on T2-weighted images (3,5,9,31). Contrast agents, particularly gadoxetic acid (Eovist and Primovist; Bayer Schering Pharma), which is excreted by hepatocytes into bile ducts, can help identify the source of the biliary leak and aid in treatment planning (32). The T1-hyperintense appearance of Eovist aids in detection of biliary leaks (32). T2-weighted MRCP additionally provides anatomic details of the biliary tree. In addition, MRCP can demonstrate the communication between the biliary ducts and suspected bilomas and often represents the primary method for biliary tree morphologic evaluation (9,10,31,32).

Finally, when available and necessary, fluoroscopic injection of existing drains can opacify the biliary system and provide diagnostic information for treatment planning (3) (Fig 4).

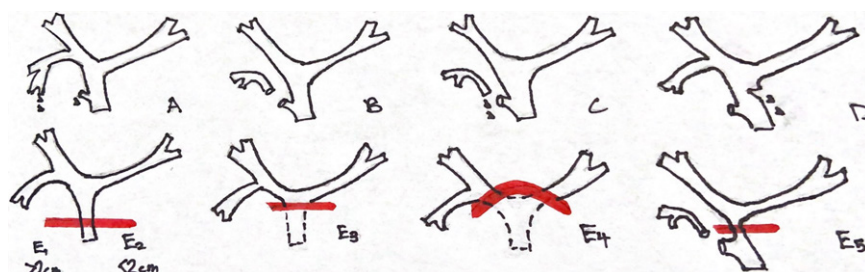


Figure 5. Bismuth-Strasberg classification of biliary injuries.

Diagnostic and Therapeutic Options

Endoscopic Retrograde Cholangiopancreatography.—ERCP is a commonly used diagnostic and therapeutic modality in the treatment of biliary leaks by the gastroenterologist in the endoscopy suite. A specially designed endoscope with a side camera and side port is guided to the duodenum, and the sphincter of Oddi is targeted and cannulated. A contrast agent is injected through the sphincter to evaluate the CBD and biliary tree to identify a biliary leak if present. Limited visualization of a peripheral injury is a limitation of ERCP, and Strasberg types C and E5 leaks may not be detected with ERCP (6). The Bismuth-Strasberg classification is described in the next section. A biliary stent can be placed if needed during the same procedure. Of note, some biliary leaks proximal to the reach of ERCP can still benefit from CBD stent placement, as the stent allows a low-flow system to divert bile away from the proximal leak (6,19).

Percutaneous Transhepatic Cholangiography.—PTC is frequently performed in patients when ERCP is not feasible due to altered surgical anatomy or in cases where antegrade access is desired to stop a leak (10). PTC is performed by using a 21- or 22-gauge Chiba needle to gain access to the biliary system and injecting a dilute contrast agent to opacify and visualize the biliary anatomy in real time (3). Often this can be used to determine the extent and location of the biliary injury. Based on this information, treatment can be planned, and various approaches are discussed below. Since the biliary system is often nondilated in patients with biliary leaks, initial PTC can be technically challenging (2,10).

Classification of Biliary Injuries

The Bismuth-Strasberg classification is commonly used to identify the location of biliary duct injuries and leaks (1,4,10,32,33) (Fig 5). The Bismuth classification was developed during the era of open surgery, with the injury localized in relation to the bile duct stump. The Strasberg system was developed after the introduction of laparoscopic surgery, forming the “E” component of injuries (1,4,6,33). The now-combined Bismuth-Strasberg classification is a widely used and helpful tool to classify the type of bile injury and guide treatment (4,6,33).

Type A is an injury of the cystic duct or minor hepatic ducts that drain the liver bed. Type B is occlusion of the biliary tree, commonly an aberrant right hepatic duct. Type C is transection without ligation of aberrant right hepatic ducts. Type D is a lateral injury to the major bile duct. Type E is an injury to

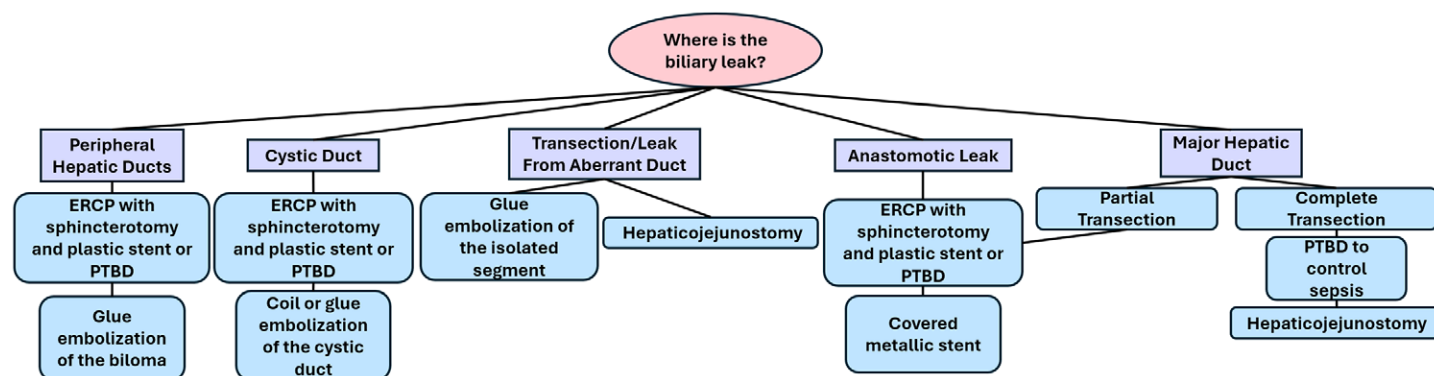


Figure 6. Diagram shows the algorithm used at the authors' institution to manage biliary leaks.

the main hepatic duct and is subdivided into five types: type E1 is an injury less than 2 cm from the confluence, type E2 is an injury more than 2 cm from the confluence, type E3 is an injury at the intact confluence, type E4 is destruction of the confluence, and type E5 involves an injury to the aberrant right hepatic duct (4,33).

There is no agreed-on universal management, and a successful outcome typically requires a multispecialty approach. It is generally thought that Strasberg A injuries can be managed by endoscopic sphincterotomy and stent placement or PTBD. Types B and C aberrant right hepatic duct injuries are traditionally treated with surgical repair, but embolization of the aberrant duct is possible in patients with complex injuries. Type D injuries can be surgically repaired or treated with retrievable covered metallic stents or a percutaneous transhepatic biliary drain. Surgical hepaticojejunostomies are the most common surgical interventions for type E injuries (6).

Treatment

Treatment of biliary leaks is unique to every patient situation. While there is no definitive consensus on the best treatment, the authors propose a treatment algorithm based on incorporation of modern minimally invasive techniques and their experience in treating biliary leaks at a tertiary liver transplant center for more than 2 decades (Fig 6). Prior to interventional treatment, medical optimization of the patient's condition with fluid resuscitation and aggressive treatment of infections and/or sepsis with broad-spectrum antibiotic therapy should be performed.

Diversion

Percutaneous Biloma Drainage.—Often, the first-line treatment performed by the interventional radiologist focuses on diversion of the bile away from the leak. This is usually done as a temporizing measure. However, this treatment can be effective in stopping the leak, secondary to reducing the biliary pressure.

If a collection is of unknown origin, then the aspiration of fluid can help provide a diagnosis (31). Biloma aspiration is commonly performed using real-time US, fluoroscopy, or CT guidance while ensuring that no major adjacent structures are injured (1,31). Depending on the size of the collection and the clinical status of the patient, a drain may be left in place

after the aspiration. A persistent and high output of bilious material after percutaneous drainage of a biloma or any collection is indicative of a biliary leak and/or fistula.

Percutaneous drains can be placed using CT, fluoroscopy, or US guidance depending on the location and complexity of the biliary leak or biloma (1). US guidance offers the operator more flexibility, real-time imaging, and Doppler US to avoid vascular structures. CT guidance provides the operator with improved details of intra-abdominal anatomy but has the drawback of radiation exposure to the patient (1,3,31). Drains may stay in place until the patient's clinical picture improves, drainage output decreases significantly (<10 mL over 24 hours), or follow-up imaging demonstrates a resolution or significant reduction in the size of the biloma. A transhepatic drain may be required to stay in place for up to 4–6 weeks to allow the tract to mature.

Endoscopic Retrograde Cholangiopancreatography.—As previously mentioned, ERCP is diagnostic and therapeutic. Once drainage of a biloma has been performed, or if clinical suspicion of a biliary leak is high, ERCP can be performed initially to identify and treat. The success rate is approximately 90% when combined with sphincterotomy (4,9,34,35). Even if a bile leak is not initially seen during the procedure, sphincterotomy and stent placement are often performed with the aim of decompressing the biliary system.

Percutaneous Transhepatic Biliary Drainage.—Biliary leaks not amenable to ERCP enter a new realm of options with individualized care on a case-by-case basis. Percutaneous transhepatic biliary drains or internal-external biliary drains can help to divert the bile and decompress the biliary tree.

As the biliary system is often nondilated, multiple punctures with a 21- or 22-gauge Chiba needle under fluoroscopic guidance are usually required. Then gentle injection of contrast agent is performed while slowly removing the needle until a bile duct is opacified. US may also be used to target triads with a similar needle withdrawal technique after placement. At this point, a diagnostic cholangiogram can be obtained followed by placement of a biliary drain. The use of general anesthesia is recommended when available. Percutaneous puncture of the gallbladder can help facilitate opacification of the biliary system to guide a percutaneous transhepatic biliary drain when

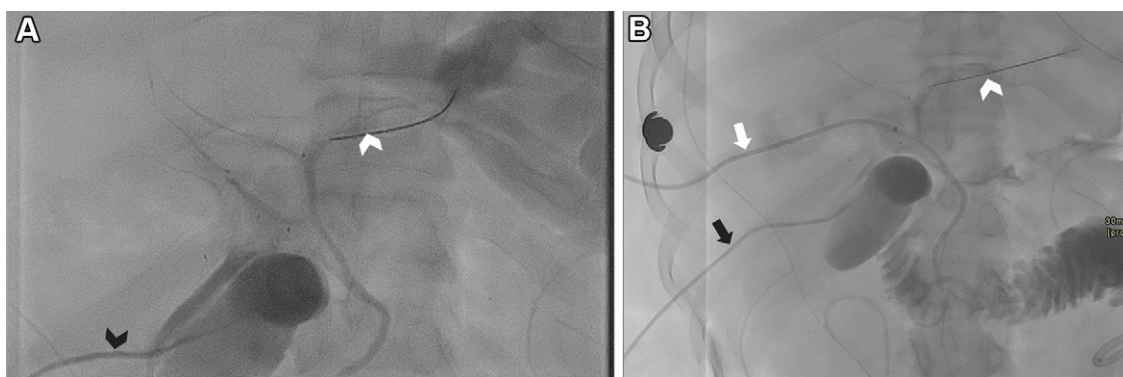


Figure 7. Biliary leak (Strasberg type A) in a 44-year-old man after a penetrating gunshot liver injury that required a percutaneous transhepatic biliary drain. **(A)** Anteroposterior cholangiogram shows that after multiple failed PTC attempts (white arrowhead in **A** and **B**), the gallbladder was punctured (black arrowhead) to allow opacification of the intrahepatic biliary ducts to guide the percutaneous transhepatic biliary drain. **(B)** Anteroposterior cholangiogram shows a right PTBD (white arrow) that was performed to divert bile away from the biliary leak. A cholecystostomy tube was also placed to further divert the bile (black arrow).

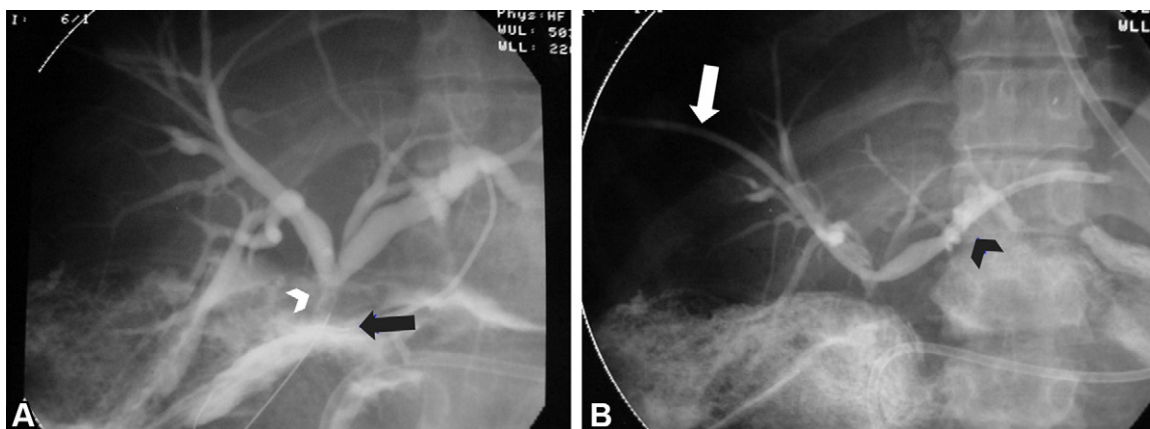


Figure 8. PTBD in a 45-year-old woman after laparoscopic cholecystectomy with an iatrogenic injury (Strasberg type E2) who presented with septic shock and peritonitis. **(A)** Anteroposterior cholangiogram of PTC shows transection just below the confluence of the left and right hepatic ducts (arrowhead) with contrast agent extravasation into the peritoneum (arrow). **(B)** Anteroposterior cholangiogram of right PTBD performed with external biliary drain placement (arrow) shows the drain extending into the left hepatic ducts (arrowhead). The patient later underwent surgical repair with hepaticojejunostomy after the sepsis was controlled.

the traditional methods have failed. The gallbladder can be filled with a contrast agent until it refluxes into the cystic duct to opacify the biliary system (Fig 7). Injecting existing surgical or percutaneous drains, when present, can help to opacify the bile ducts and identify the source of the leak (2,3).

During the initial procedure, crossing a leak with an internal-external drain is preferred but not always possible, such as in cases of complete ductal transection. In this scenario, external biliary drainage can be performed to begin treatment. This can be followed with placement of an internal-external drain after the acute process has resolved. PTBD serves as a temporary solution and can also help guide future surgical reconstruction (6) (Figs 8, 9). After maturation of the PTBD access tract, which typically requires 3–4 weeks, the access site can be used to perform advanced procedures.

Sealing Biliary Leaks

After a period that varies from weeks to months, a bile leak is expected to resolve through the natural healing processes of

the body. If a biliary leak persists despite diversion and conservative management, then additional methods of embolization can be considered for definitive management. These are dependent on the patients' anatomy, site of the leak, alternative surgical and endoscopic options, and patient preference.

Biliary Embolization.—Biliary embolization is often performed in two scenarios. The first scenario is to occlude the injured isolated duct in the setting of a fistula, and the second is to occlude an entire biliary segment for induction of hepatic segmental atrophy (8). There are many embolization materials such as absolute alcohol, acetic acid, and cyanoacrylate (8). *N*-butyl cyanoacrylate (NBCA), which is used at the authors' institution, is a low-viscosity tissue glue monomer that polymerizes and becomes solid when in contact with fluids at a neutral pH (2). Due to the viscosity of the glue, a combination of glue and coils is sometimes preferred, with coils used as scaffolding for the glue (2). NBCA is commonly mixed with ethiodized oil to improve radiopacity and control the polymerization rates.

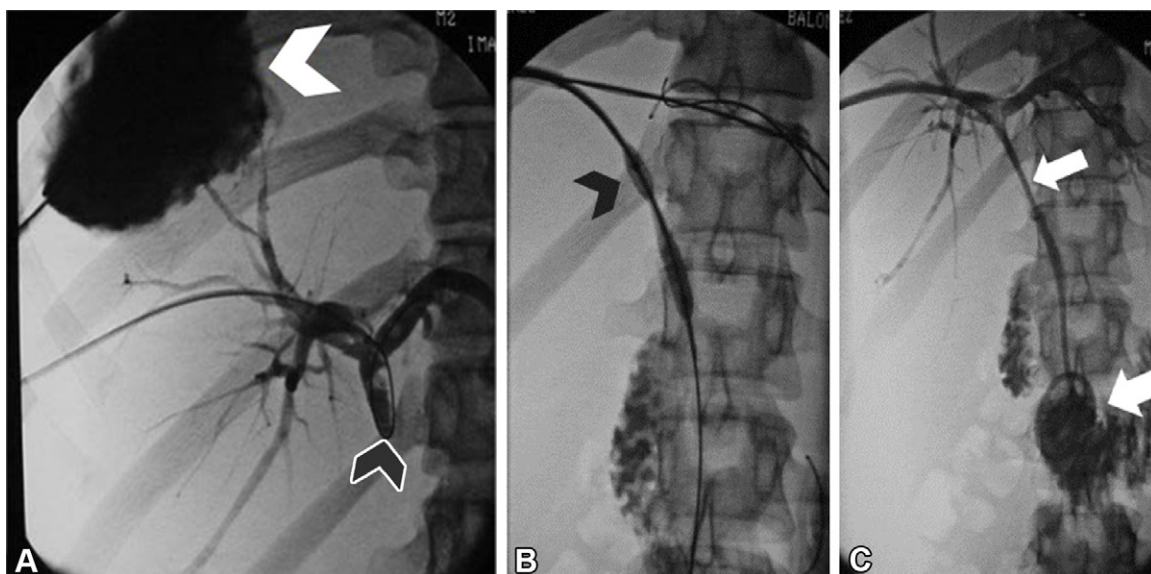


Figure 9. Iatrogenic injury (Strasberg type E1) in a 27-year-old woman with a large biloma and persistent high output of bile through the drain after laparoscopic cholecystectomy. **(A)** Anteroposterior cholangiogram with contrast agent injected through a drain shows the biloma (white arrowhead) communicating with the biliary system and CBD occlusion (black arrowhead). **(B, C)** Anteroposterior cholangiograms with percutaneous access show that the CBD occlusion was crossed with a wire and balloon dilation (arrowhead in **B**) was performed with PTBD (arrows in **C**). The patient later underwent surgical repair with hepaticojejunostomy.



Figure 10. Stent placement in a 31-year-old woman with a high-output biliary leak for 3 months (Strasberg type A) after laparoscopic cholecystectomy. The biliary leak did not respond to ERCP stent placement. Anteroposterior cholangiogram with contrast agent injected through a drain shows communication with an isolated segment 5 radicle (arrows) without opacification of the main biliary ducts. The CBD stent (arrowhead) was successfully glued.

Embolization of biliary fistulas is usually reserved for leaks that are not controlled with diversion after PTBD or endoscopic biliary stent placement. Iatrogenic injuries of isolated or orphaned biliary radicles after partial hepatectomies or liver ablation can be safely and effectively treated with glue embolization (Figs 10–12). To perform glue embolization, the biliary ducts are usually selected using an existing percutaneous transhepatic biliary drain access, if

present, or another drainage tract such as a biloma drain. Depending on the size and tortuosity of the ducts, selective catheterization with a 4F to 5F base catheter or a microcatheter is performed. After the system is flushed with a 5% dextrose solution, a mixture of NBCA and ethiodized oil, usually with a 1:1 or 1:2 ratio, is injected through the microcatheter until a cast of glue fills the bile ducts.

The percutaneous tract can also be embolized with glue. When the connection between the biliary ducts and the biloma is too small to be catheterized, glue embolization of the biloma itself can be performed with care to avoid glue migration into the bile ducts (Fig 11). To ensure adequate glue penetration in high-output leaks, balloon occlusion of the outflow tract can be performed followed by an injection of a mixture of lipiodol and glue (NBCA) (Fig 12). Of note, if the sclerosing material leaks into a well-innervated surface, such as the peritoneum, severe pain can occur (8). The use of an existing mature tract or placement of a sheath in the tract may help to prevent this issue.

In cases of chronic biliary-cutaneous fistulas, a combination of plugs or coils and glue is usually used. Plugs or coils are first placed near the entry to the bile ducts to function as scaffolding and to partially reduce flow, and then the tract is embolized with glue (Fig 13).

The use of a plug, with a membrane that acts as scaffolding, combined with NBCA glue has been described to close a biliopleural fistula (30) and more recently in seven cases of a variety of iatrogenic biliary leaks (36). At their institution, the authors have used NBCA glue with a plug as scaffolding to successfully seal a biliary fistula after TACE (Fig 14).

Coil Embolization of the Cystic Duct.—After surgical removal of the gallbladder, biliary leaks that originate from the cystic duct can be due to failure to ligate the cystic duct or

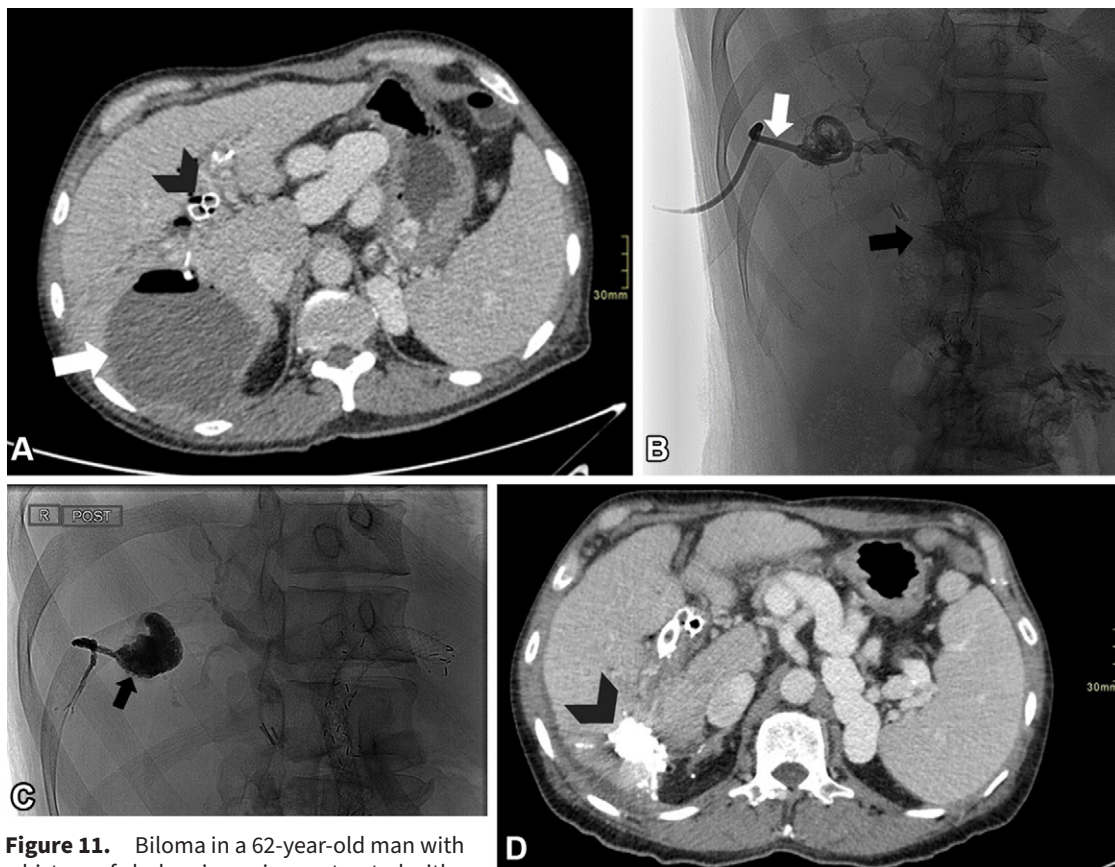


Figure 11. Biloma in a 62-year-old man with a history of cholangiocarcinoma treated with chemotherapy and kissing metallic stents. The patient underwent ablation of a 2-cm right lobe tumor 4 weeks previously and presented with abdominal pain and high fever. **(A)** Axial contrast-enhanced CT scan shows a large infected biloma (arrow). Note the metallic biliary stents (arrowhead). After drainage of the biloma, there was high-output biliary drainage that persisted for 3 months. **(B)** Anteroposterior cholangiogram with contrast agent injected through a drain (white arrow) shows communication of the cavity with the right biliary ducts and the stent (black arrow). There was an attempt to cannulate the bile ducts through the biloma cavity, which failed. **(C)** Anteroposterior cholangiogram shows that the cavity and tract were filled with glue (arrow), and the leak resolved. **(D)** CT scan 3 months later shows resolution of the biloma with residual glue cast in the area (arrowhead).

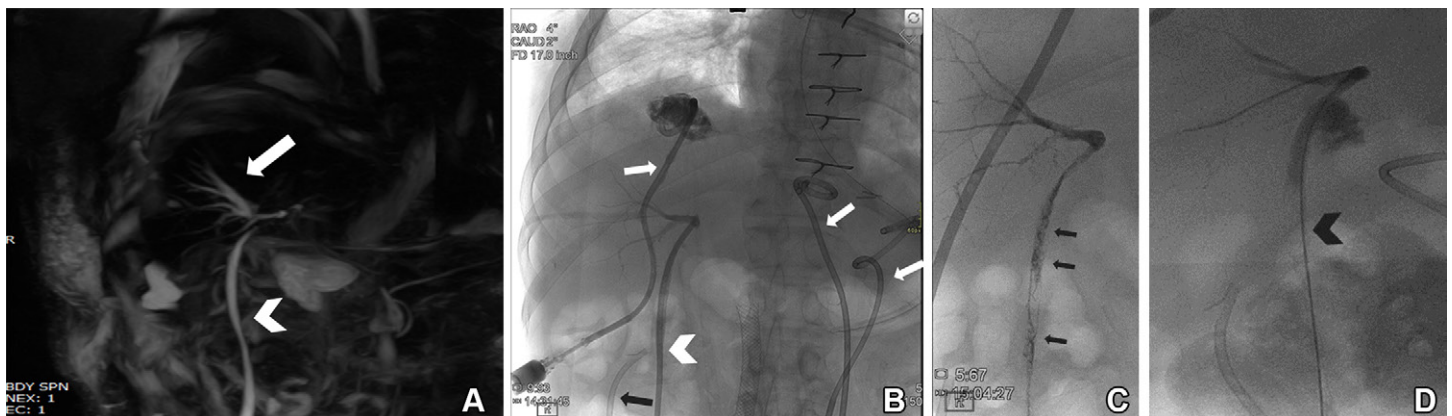


Figure 12. CBD injury (Strasberg type E2) in a 62-year-old man after laparoscopic cholecystectomy with T-tube placement. The patient had the T-tube accidentally removed, resulting in a CBD tear that led to a biloma and multiple abscesses. **(A)** MR cholangiogram shows the isolated bile ducts (arrow) in segment 5 and the leak into the drain (arrowhead). **(B)** Anteroposterior cholangiogram with contrast agent injected through a drain shows the biloma and multiple perihepatic abscesses drained with pigtail catheters (arrows). An external biliary drain (arrowhead) was placed in the isolated bile duct with control of the biloma. Note a plastic stent in the CBD that was placed using ERCP (black arrow). After the drain was removed, the patient developed a biliary cutaneous fistula. **(C)** Anteroposterior cholangiogram with contrast agent injection shows an attempt to control the biliary cutaneous fistula tract with glue injected into the tract (arrows). **(D)** Anteroposterior cholangiogram 1 week later demonstrates fistula tract recannulation with a wire and catheter (arrowhead). Balloon occlusion was performed in the outlet, and a mixture of lipiodol, histoacryl, and ethanol was injected into the right hepatic duct. The biliary fistula resolved.

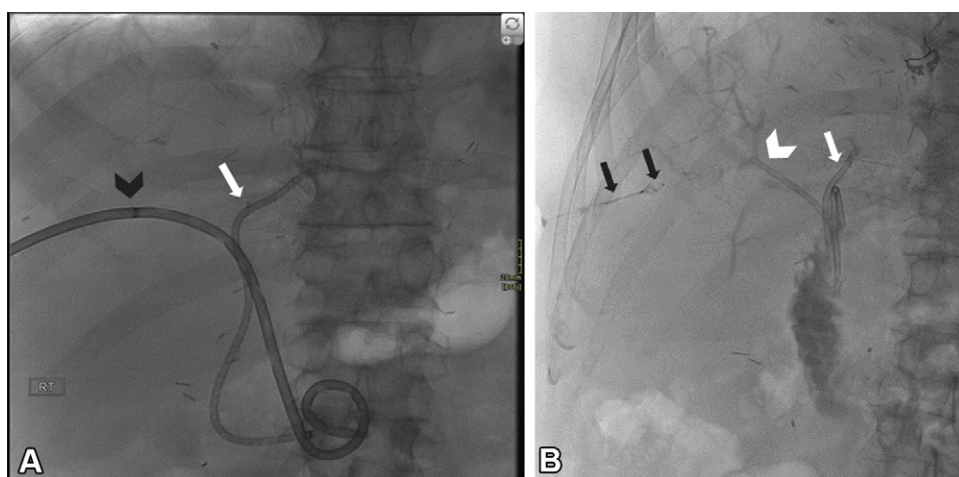


Figure 13. Biliary necrosis in a 59-year-old man who underwent orthotopic liver transplant with retransplant 1 year later secondary to hepatic artery thrombosis. **(A)** Anteroposterior cholangiogram shows a long-term percutaneous transhepatic biliary drain (arrowhead) placed with articulated plastic stents (arrow). **(B)** Anteroposterior cholangiogram shows that the stents (white arrow) were converted to internal stents (arrowhead) with ERCP 5 years after the transplant. A biliary cutaneous fistula formed through the old percutaneous tract, which was closed using an Amplatzer (Abbott) plug and glue (black arrows). The biliary leak resolved.

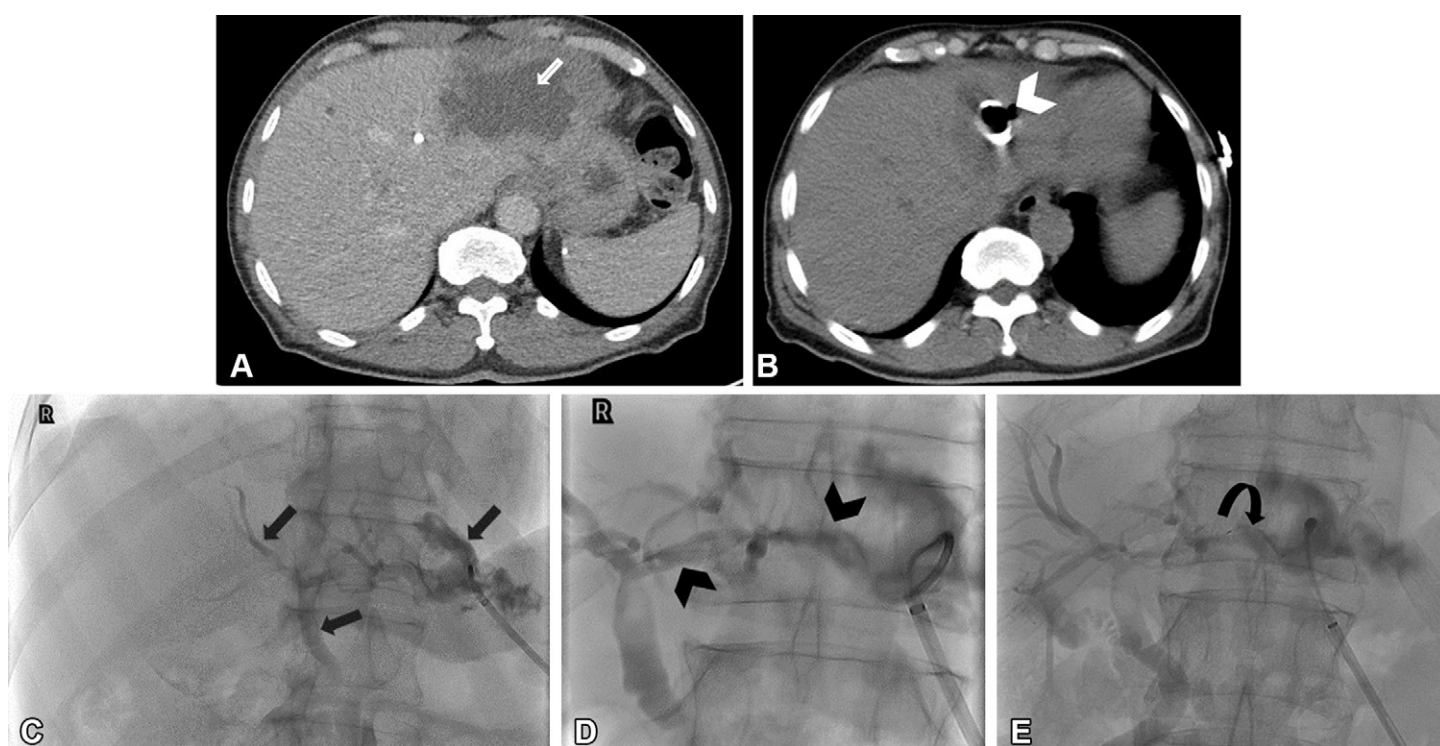


Figure 14. Biliary fistula in a 78-year-old man who underwent TACE of a left lobe hepatocellular carcinoma and presented with fever and sepsis afterward. **(A)** Axial CT scan shows a large left lobe biloma (arrow) that was drained. **(B)** Axial CT scan shows the drain (arrowhead) terminating in the biloma cavity. The drainage had a high output of bile for several weeks afterward. **(C)** Anteroposterior cholangiogram with contrast agent injected through a drain shows communication between the biloma and left biliary system (arrows). **(D)** Anteroposterior cholangiogram shows that glue embolization (arrowheads) was attempted but failed. **(E)** Anteroposterior cholangiogram of a second procedure where the biliary leak was cannulated shows placement of a plug device (arrow) (Caterpillar Arterial Embolization Device; Becton Dickinson), followed by glue embolization. The leak stopped, and the drain was later removed.

incomplete ligation, specifically when a partial cholecystectomy is performed. When diversion with ERCP stent placement and sphincterotomy fails, cases are typically amendable to cystic duct embolization.

Cystic duct stump coil embolization is a minimally invasive, safe, cost-effective procedure that can treat a leaking cystic duct stump in selected patients (2). Access into the cystic duct stump can be obtained via a surgical drain, which is removed over a guidewire and replaced with a catheter, allowing targeted cystic duct cannulation (Fig 15). Coil embolization is performed using the base catheter.

Additionally, the cystic duct can be cannulated using a catheter and guidewire combination from a transhepatic approach using an existing percutaneous transhepatic biliary drain access. This approach can be difficult given the acute angulation of the cystic duct, but coils can be placed with good success (Figs 16, 17). The cystic duct can alternatively be embolized with glue (NBCA or Onyx; Medtronic) if there is an inadequate length for safe coil deployment (2). Special care is needed to avoid coil or glue migration into the CBD (31), as these foreign bodies could lead to stenosis, cholangitis, or stone formation.

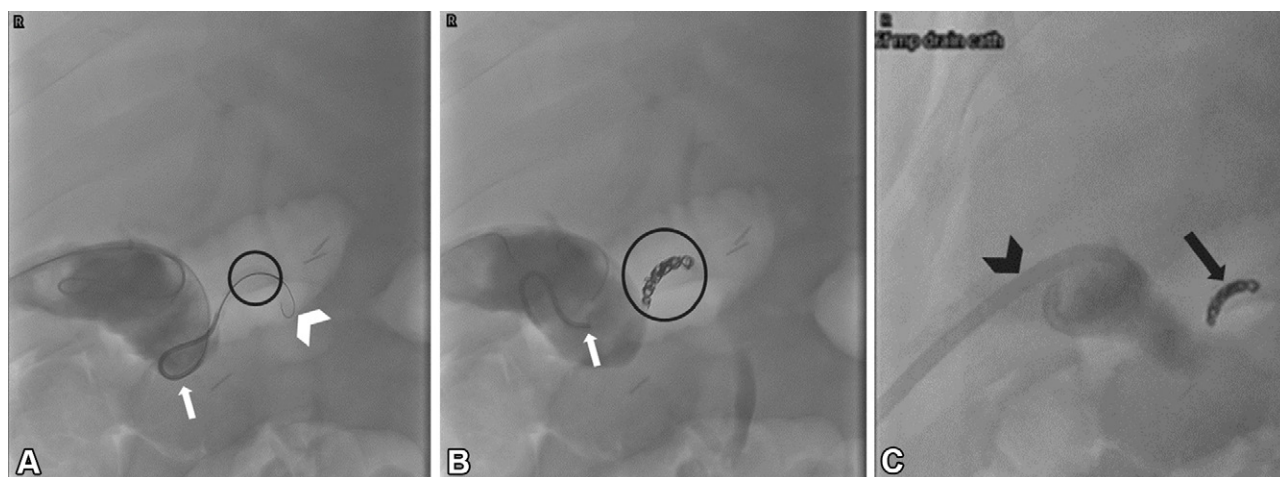


Figure 15. Coil placement in a 47-year-old man with a history of Roux-en-Y gastric bypass 12 years earlier. The patient also underwent a partial cholecystectomy for acute cholecystitis 2 weeks previously. The patient experienced greater than 1000 mm/d of bilious output in the postoperative drain, secondary to a Strasberg type C injury. **(A)** Anteroposterior cholangiogram shows a catheter (Sos Omni Selective; AngioDynamics) (arrowhead) that was introduced through the existing drain tract (arrow); the cystic duct was cannulated with a microcatheter (circle). **(B)** Anteroposterior cholangiogram shows that once the position was confirmed, coils were placed (circle) as the microcatheter (arrow) was removed from the cystic duct. **(C)** Anteroposterior cholangiogram shows the deployed coils (arrow) and an external drain (arrowhead). The output ceased after the procedure.

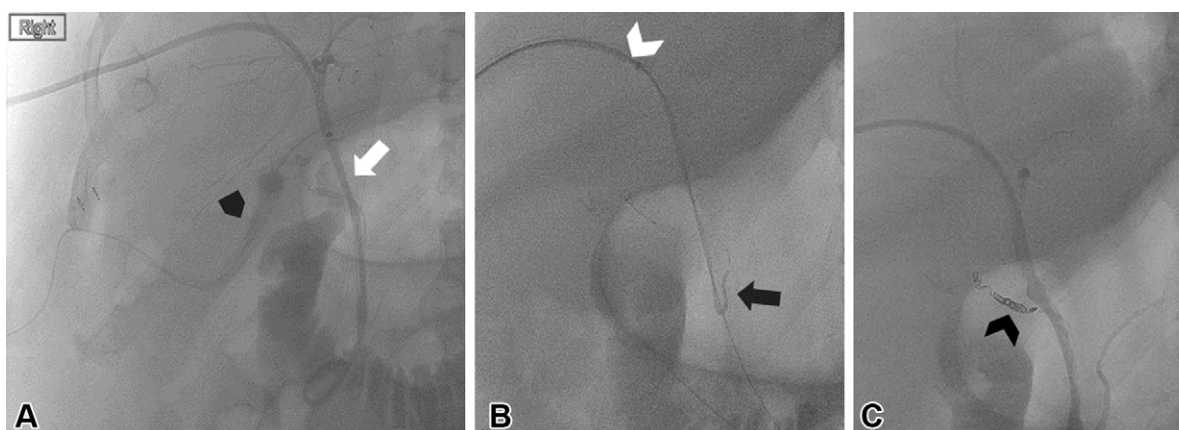


Figure 16. Coil placement in a 63-year-old man with a leaking cystic duct stump (Strasberg type A) seen 6 weeks after laparoscopic cholecystectomy for continued high output in the surgical drain, which was confirmed as a biliary leak using a HIDA scan. **(A)** Anteroposterior cholangiogram shows a percutaneous transhepatic biliary drain that was initially placed (arrow). Note leakage of contrast agent into an existing surgical drain (arrowhead). **(B)** Anteroposterior cholangiogram shows the sharp angulation of the cystic duct, which resulted in a difficult cannulation. However, a 4F catheter (arrow) (Sos Omni Selective; AngioDynamics) was placed inside an 8F sheath (arrowhead) to cannulate the cystic duct. **(C)** Anteroposterior cholangiogram shows that after the cystic duct was cannulated with a microcatheter, multiple coils (arrowhead) were placed to seal the fistula.

Biliary Stents.—Covered self-expanding metallic stents (SEMS) have been used successfully to seal biliary leaks, especially after liver transplant and when iatrogenic injuries occur after laparoscopic cholecystectomy or other biliary surgeries (4). (Figs 11, 13, 17). SEMS can be placed using ERCP or, less frequently, a PTBD access. Plastic stents are an option in addition to metallic stents (35,37). The use of covered SEMS has the dual purpose of sealing the leaks and treating the disruption or stenosis of the bile ducts.

Surgical Treatment of Biliary Leaks

There are extensive options regarding surgical management of biliary leaks beyond the scope of this article. After recog-

nition of Strasberg type B through type E injuries, primary surgical repair of the leak is typically performed, but effective nonsurgical options are also available (Fig 6). Many of the surgical treatment options consist of suturing the leak, if visible, or diverting the bile flow by anastomosing a mobilized loop of jejunum (commonly Roux-en-Y hepaticojejunostomy) to the biliary system (1,31,38). For complex biliary leaks, surgery may be performed after conservative or less invasive therapies have failed. These surgeries have a risk of high morbidity (38%–47%) and mortality (2%–9%), with the most common long-term complication of a Roux-en-Y hepaticojejunostomy being anastomotic stricture in 10%–20% of patients at 11–30 months after surgery (1,3,6,10,11,13,31). As previously

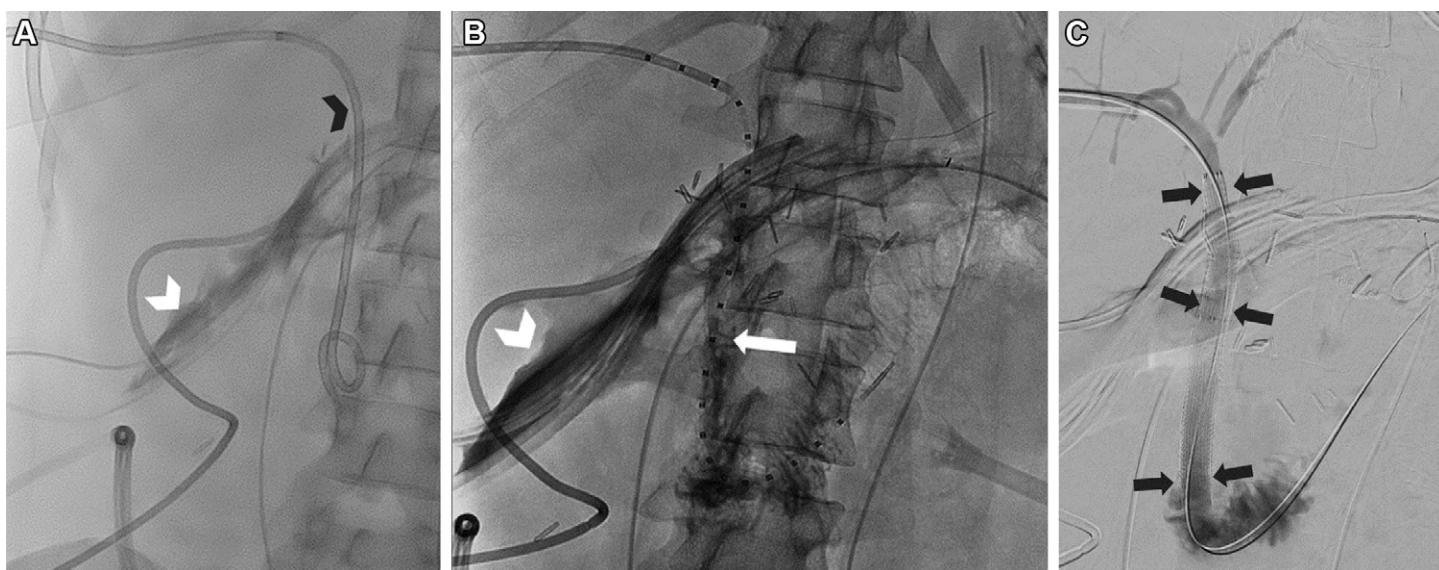


Figure 17. Coil placement in a 29-year-old man who underwent hepaticojejunostomy reconstruction for a stab wound, with multiple failed biliary-enteric anastomoses. The patient had increasing output in surgical drains up to 1.7 L. **(A)** Initial anteroposterior cholangiogram obtained with placement of an internal-external drain (black arrowhead) shows an anastomotic leak and contrast agent extravasation (white arrowhead in **A** and **B**). **(B)** Subsequent anteroposterior cholangiogram shows that a marker catheter was used to measure the length of the injury (arrow). **(C)** Anteroposterior digital subtraction cholangiogram shows overlapping covered stents (arrows) (Viabhan; W L Gore) from the hepatic duct to the jejunum (arrows). An external drain was placed at the end of the case, and the leak resolved.

described, percutaneous drainage of infected bilomas and PTBD are especially useful techniques to provide source control. The diversion of bile allows the clinical condition of patients to improve, and they can later undergo more definitive surgical repairs. PTBD is also especially useful to help identify the bile ducts during surgery.

Complications

PTBD is a very safe procedure when performed with careful technique and imaging guidance. Major complications have an estimated incidence of 2% in PTC and 2.5% in PTBD, with an overall rate of 3%–10% (1,6,39). Major complications include but are not limited to sepsis, cholangitis, bile leak, and vascular injury, resulting in hemobilia, hemoperitoneum, hematoma, adjacent organ injury, and pleural complications (0.8%–2.1%) (1,6,28,39). Tube migration, bleeding, and accidental transgression of bowel or pleura are rare complications.

Infectious complications such as cholangitis and sepsis are well known, and broad-spectrum antibiotic therapy is recommended with adequate coverage of gram-negative and anaerobic bacteria (40). At the authors' institution, the published *Journal of Vascular and Interventional Radiology* guidelines for prophylaxis are followed by using 1–2 g of ceftriaxone, 4.5 g of piperacillin and tazobactam, or 1.5–3 g of ampicillin-sulbactam intravenously in patients who have undergone liver transplant or high-risk patients. These guidelines are used prior to any biliary interventions, including treatment of biliary leaks (40).

For the embolization of biliary leaks or the cystic duct, the most feared complication is migration of glue or coils into the central biliary ducts. As previously mentioned, glue or coils can serve as a nidus of infection, or stone formation or could create strictures of the bile ducts. Migrated embolic material can be retrieved with snares or baskets or can be pushed into

the small bowel using Fogarty balloons. Late migration of the coil was reported in one of seven patients after cystic duct embolization in one study (2).

Outcomes

The success rates for biliary drainage of small leaks, cystic duct leaks, and traumatic leaks exceed 90% without the need for subsequent surgery (41). ERCP techniques are highly effective for most postcholecystectomy leaks from the cystic duct and duct of Luschka but for only 40% of leaks at the other locations (42).

There are multiple small series and case reports on the successful embolization of refractory biliary leaks. To the best of our knowledge, there are no large series published given the rarity of complex biliary leaks refractory to initial diversion. A recent series of seven patients with iatrogenic biliary leaks who were treated with a polytetrafluoroethylene-covered microplug reported a technical and clinical success of 100% (36). Nezami et al (2) reported that coil embolization of the cystic duct in seven patients resulted in successful exclusion of the leak in all patients and late migration of the coil into the CBD after 2.5 years in one patient. In most of our cases, cessation of biliary leaks was experienced immediately after intervention or within a few weeks after the intervention.

Conclusion

Biliary leaks can manifest in many ways and are associated with significant morbidity and mortality, especially if diagnosis and intervention are delayed. There are many options to address biliary leaks, including biloma drainage, PTBD, ERCP, embolization, and placement of plastic or metallic stents. A multidisciplinary approach is needed to tailor the most effective and appropriate treatment method for each specific case.

Multiple techniques are sometimes needed over an extended time period for treatment of complex biliary leaks. Interventional radiology plays an ever-increasing role in nonoperative management of biliary leaks.

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